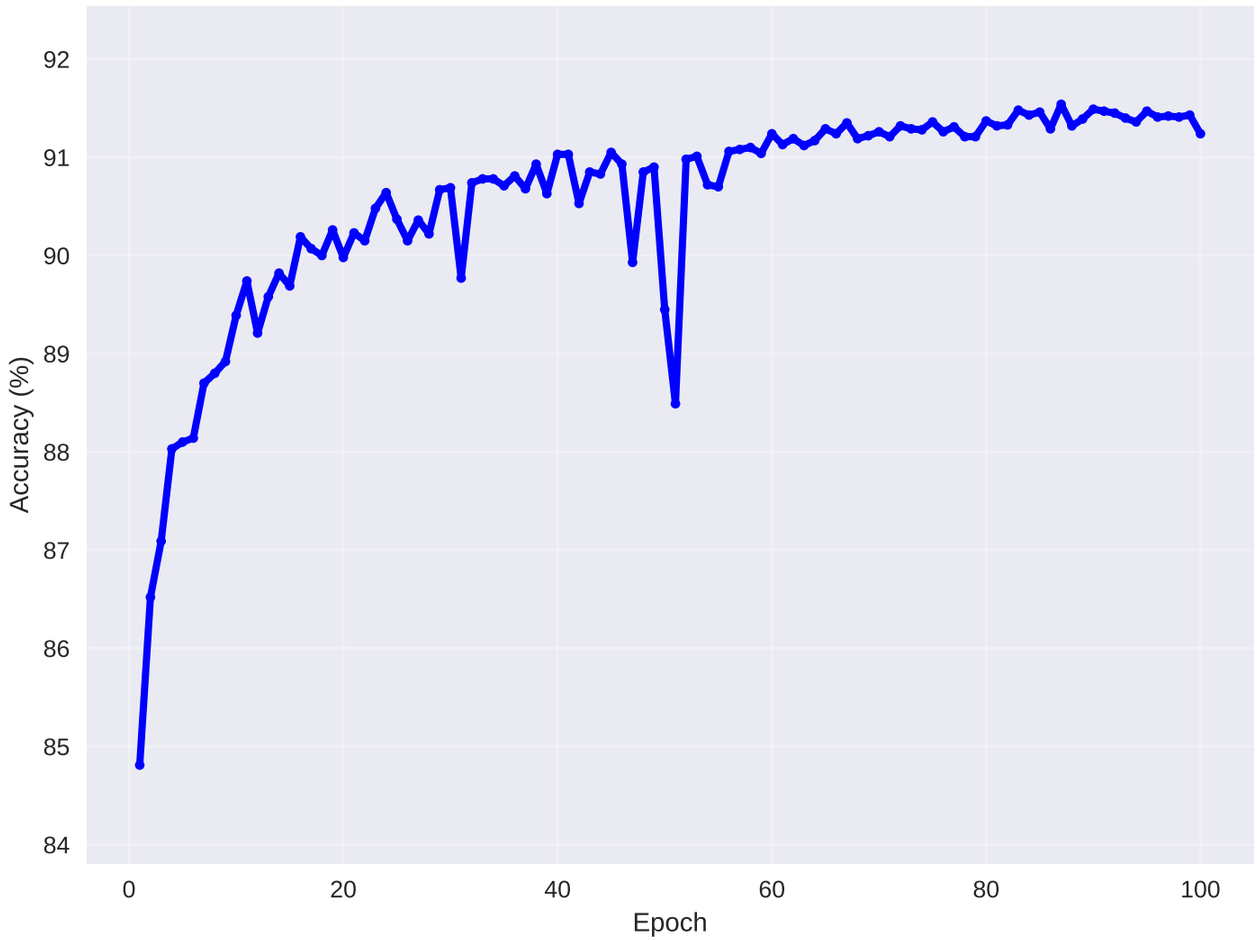
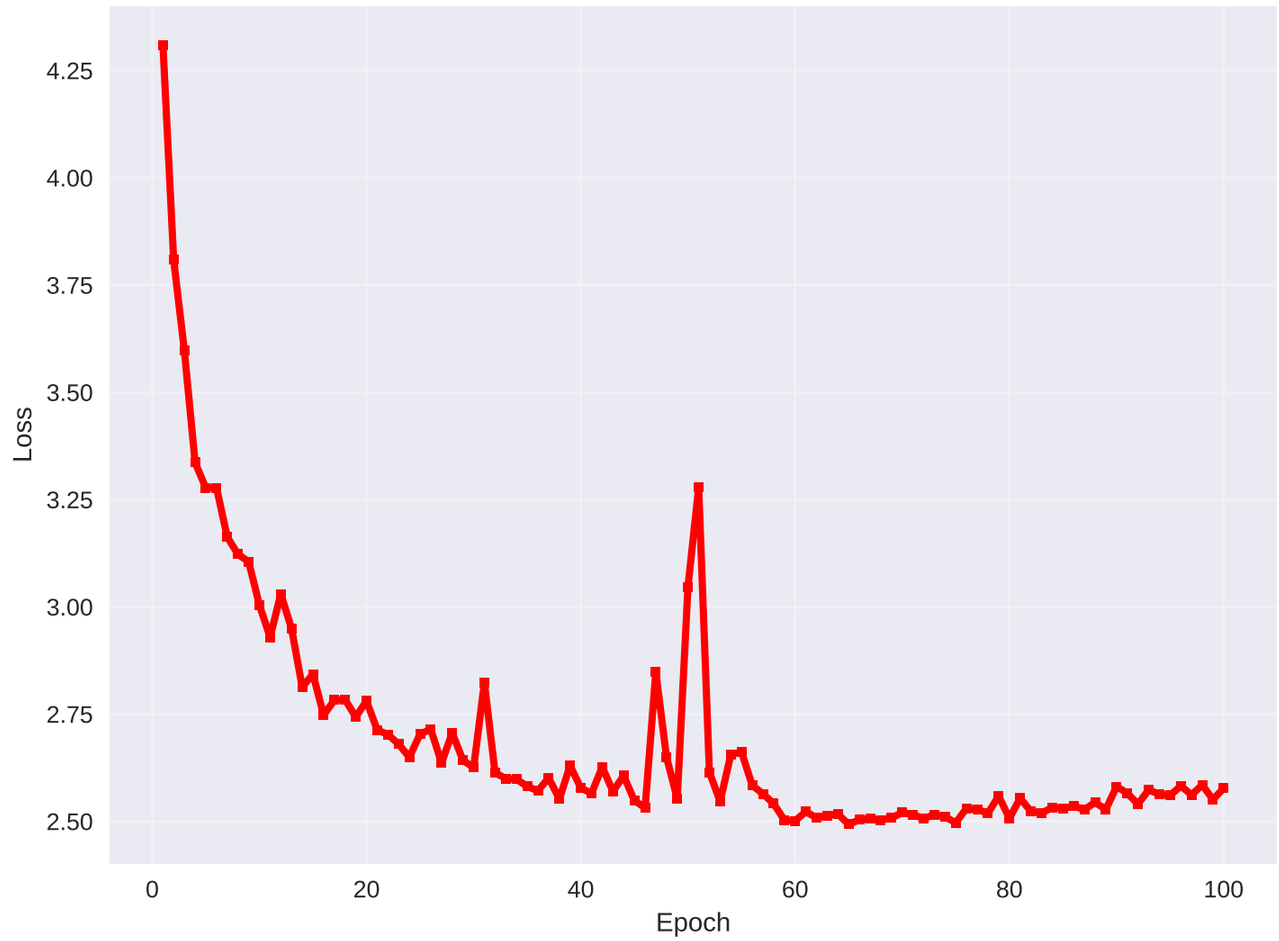
Federated Learning - Main Metrics Overview

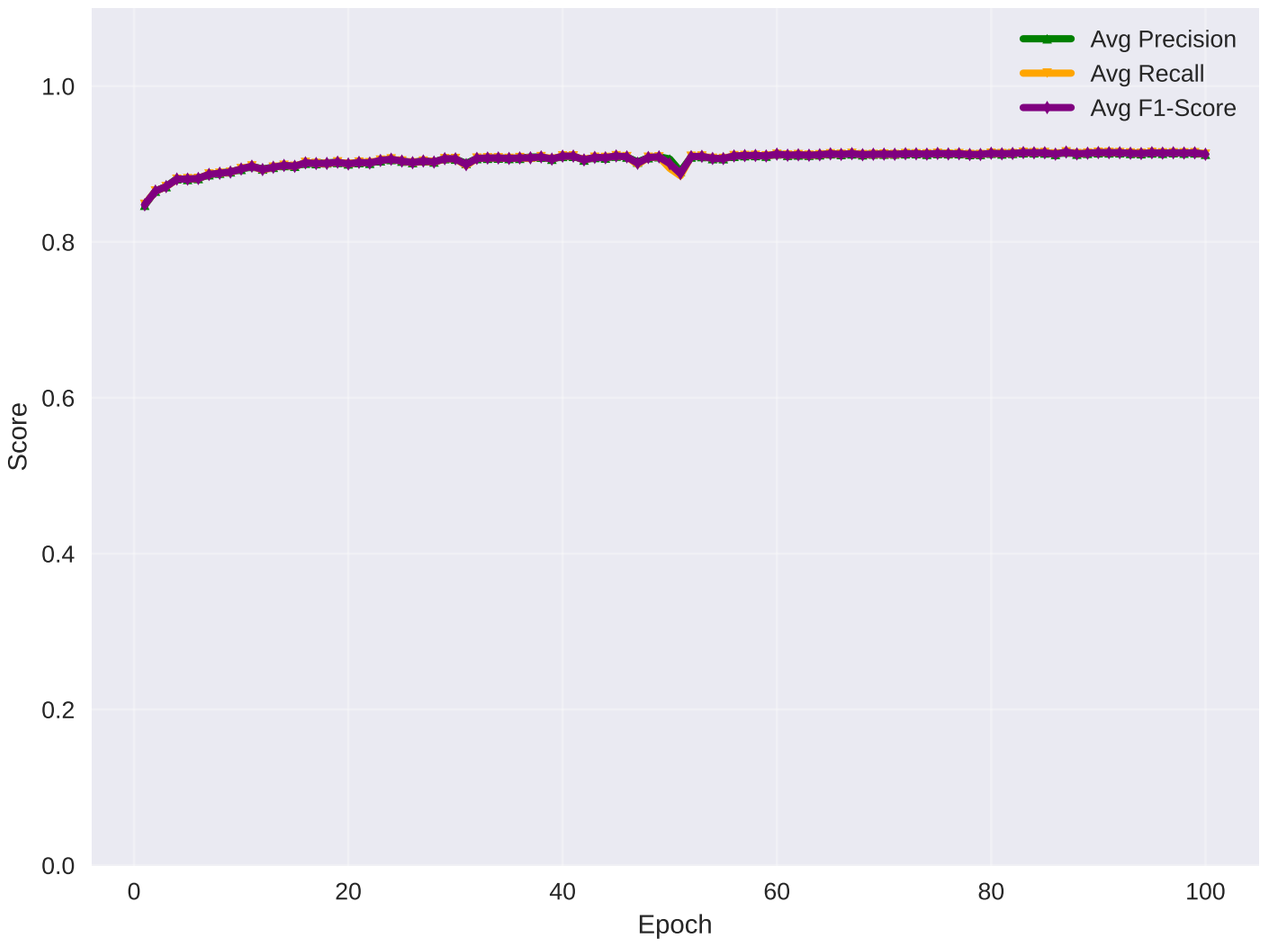
 Global Model Accuracy Evolution



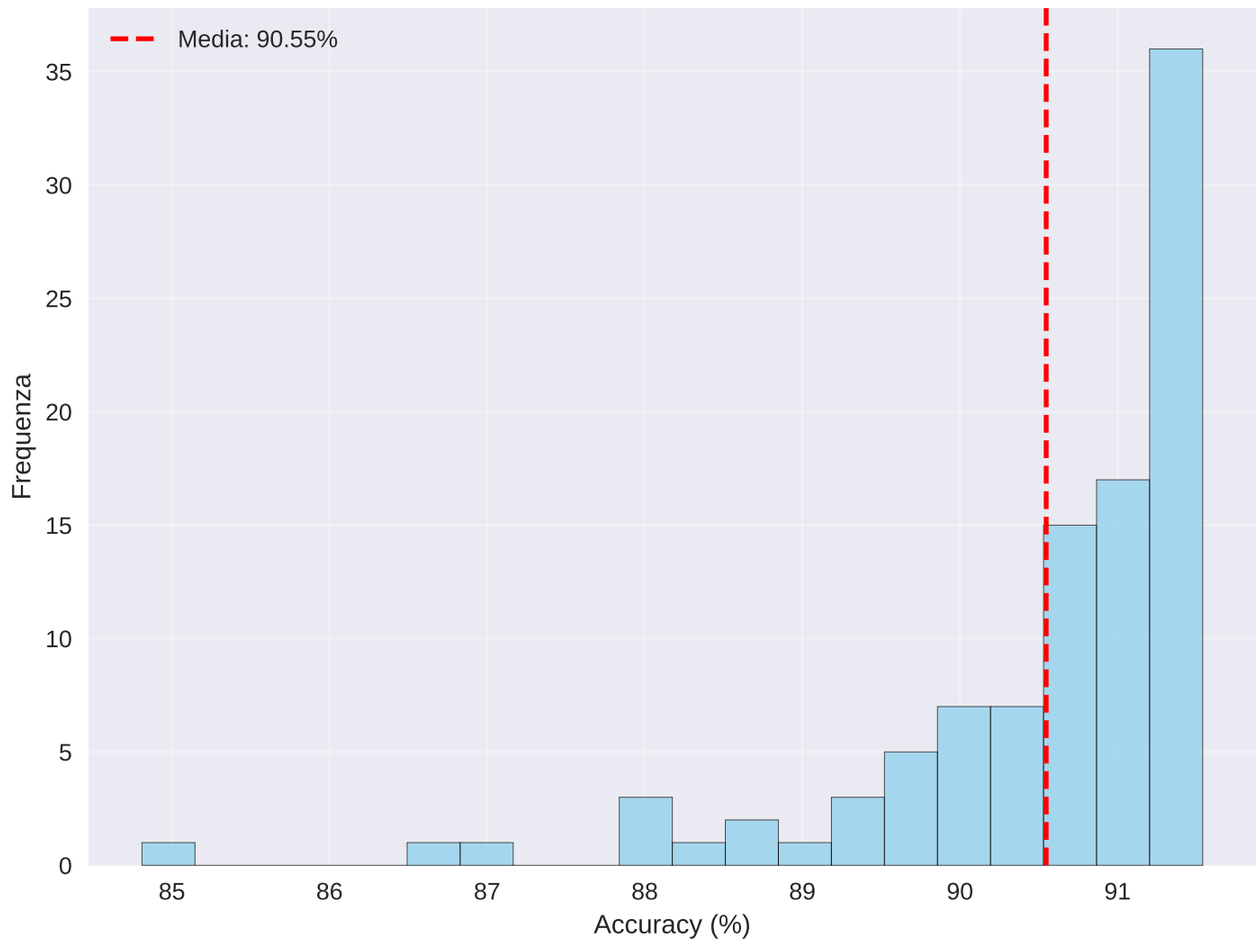
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

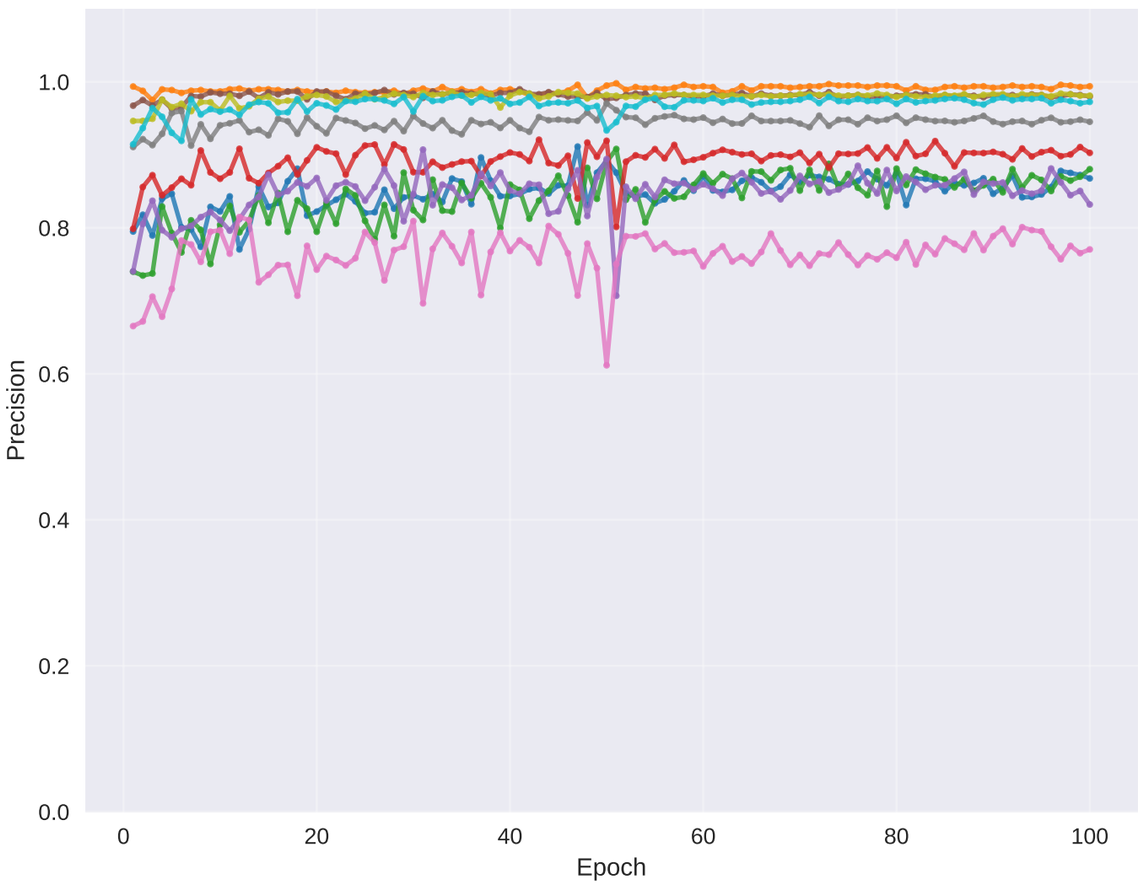


 Distribuzione Accuracy

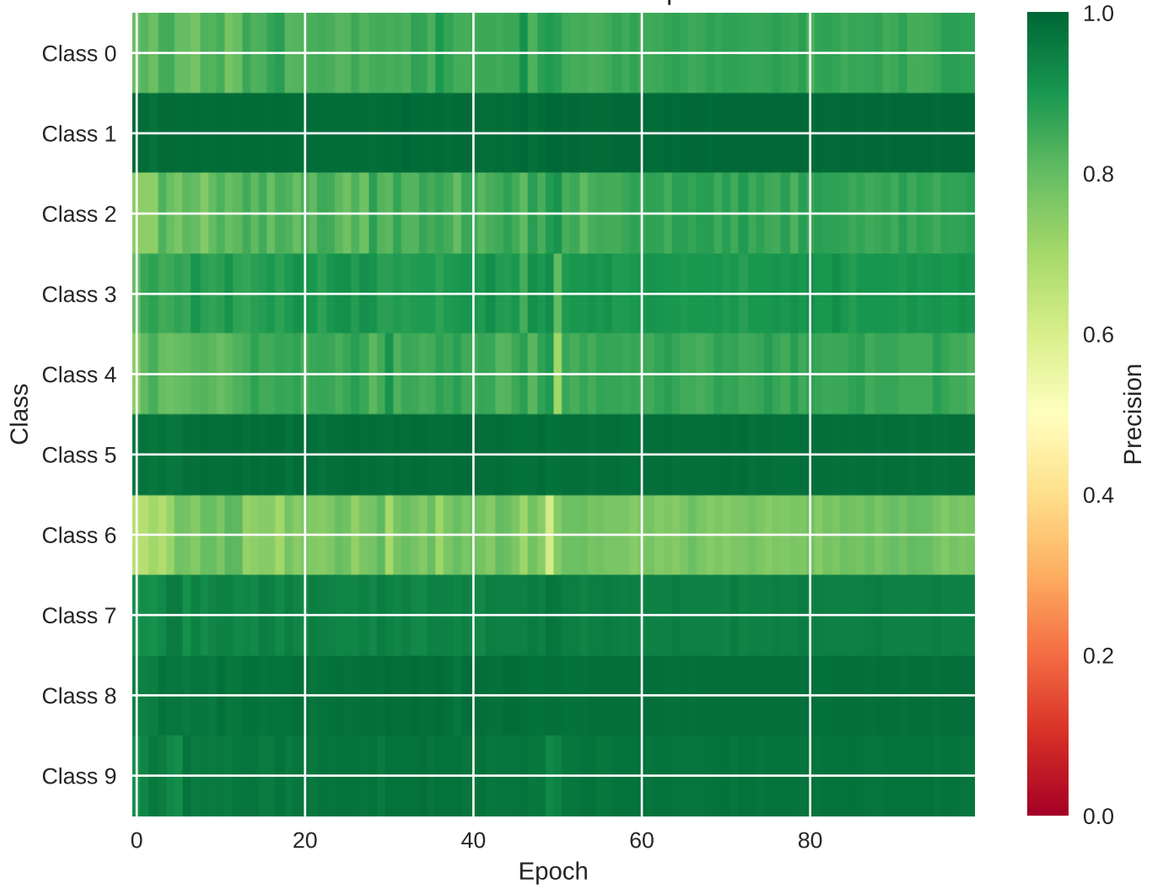


Per-Class Precision Analysis

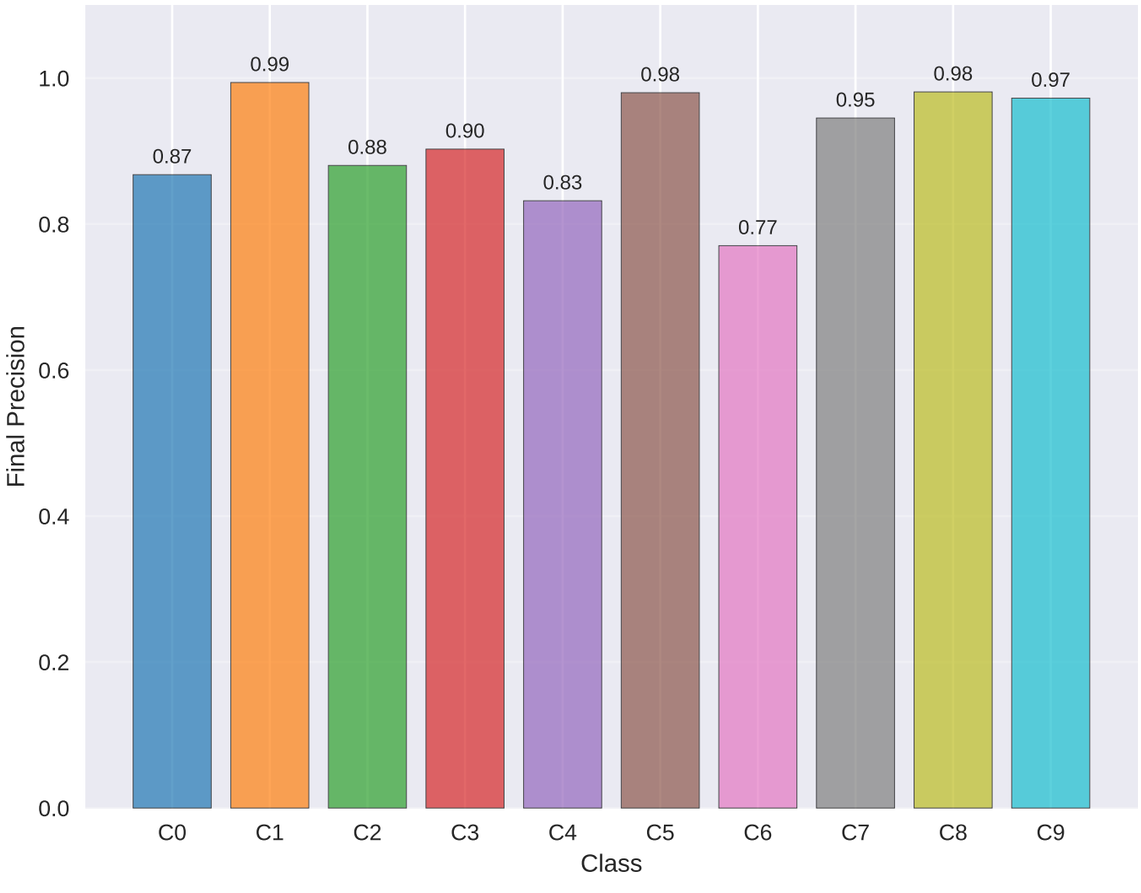
□ Per-Class Precision Evolution



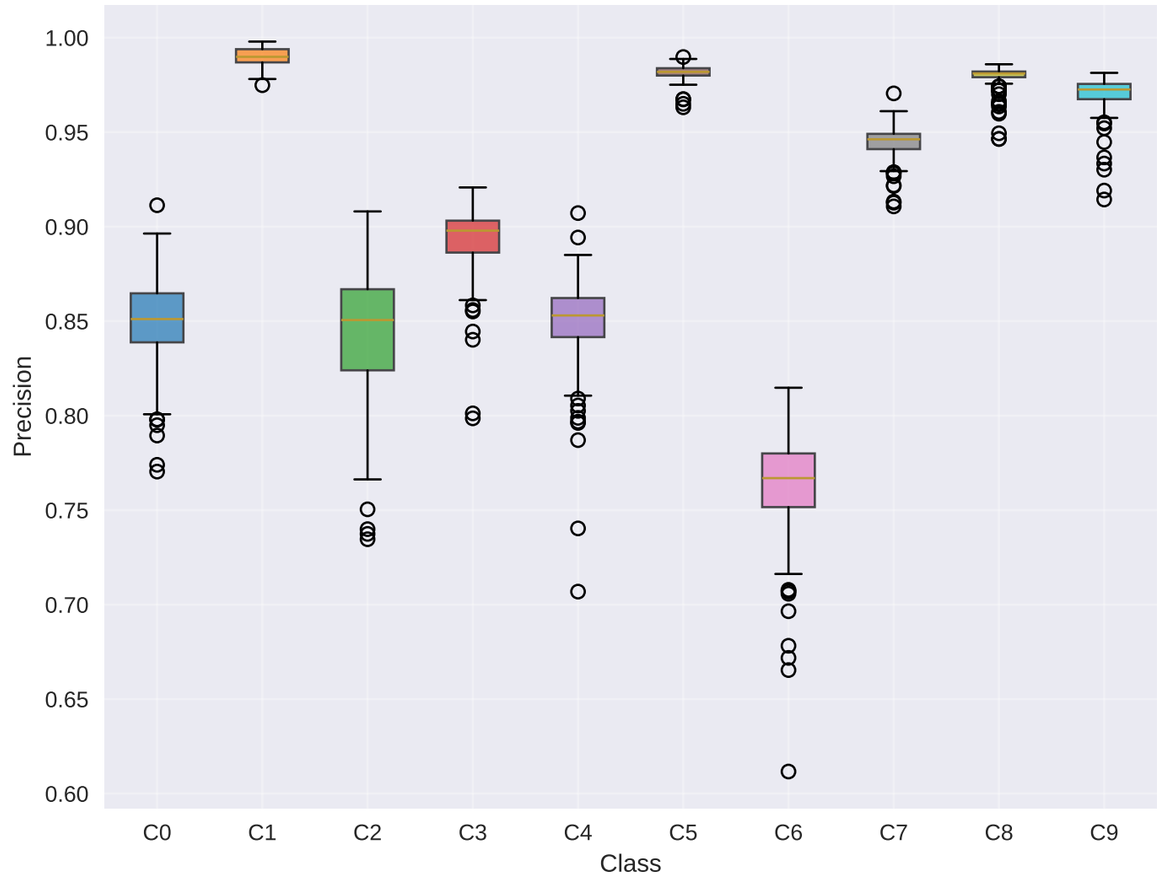
□ Precision Heatmap



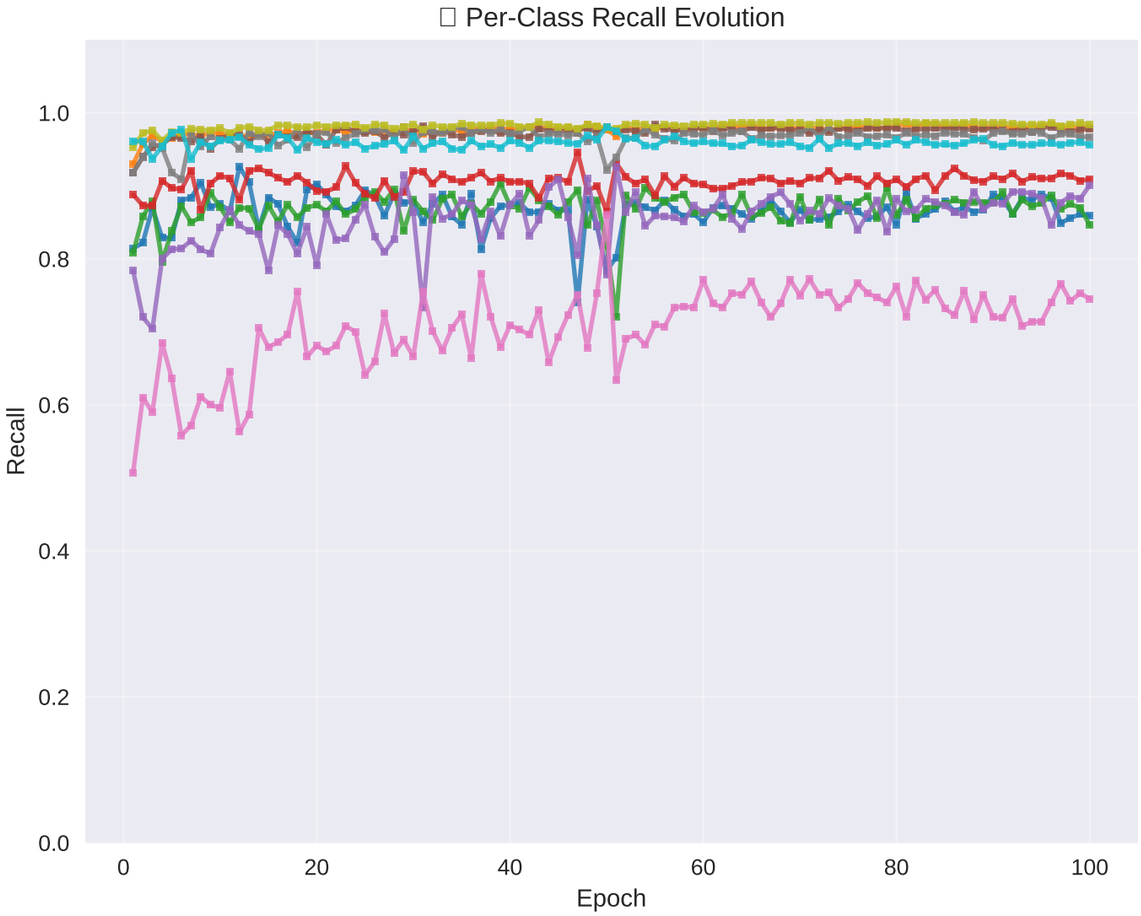
□ Final Precision per Class

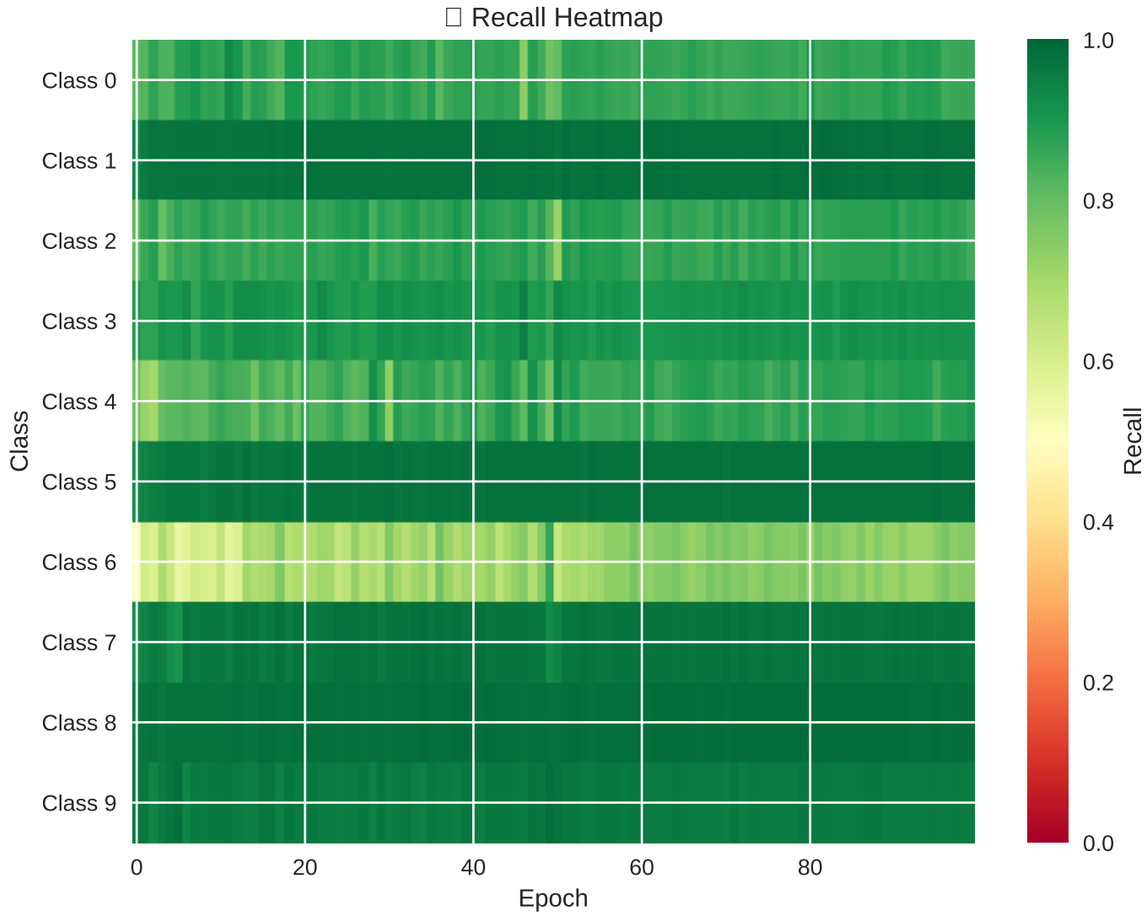


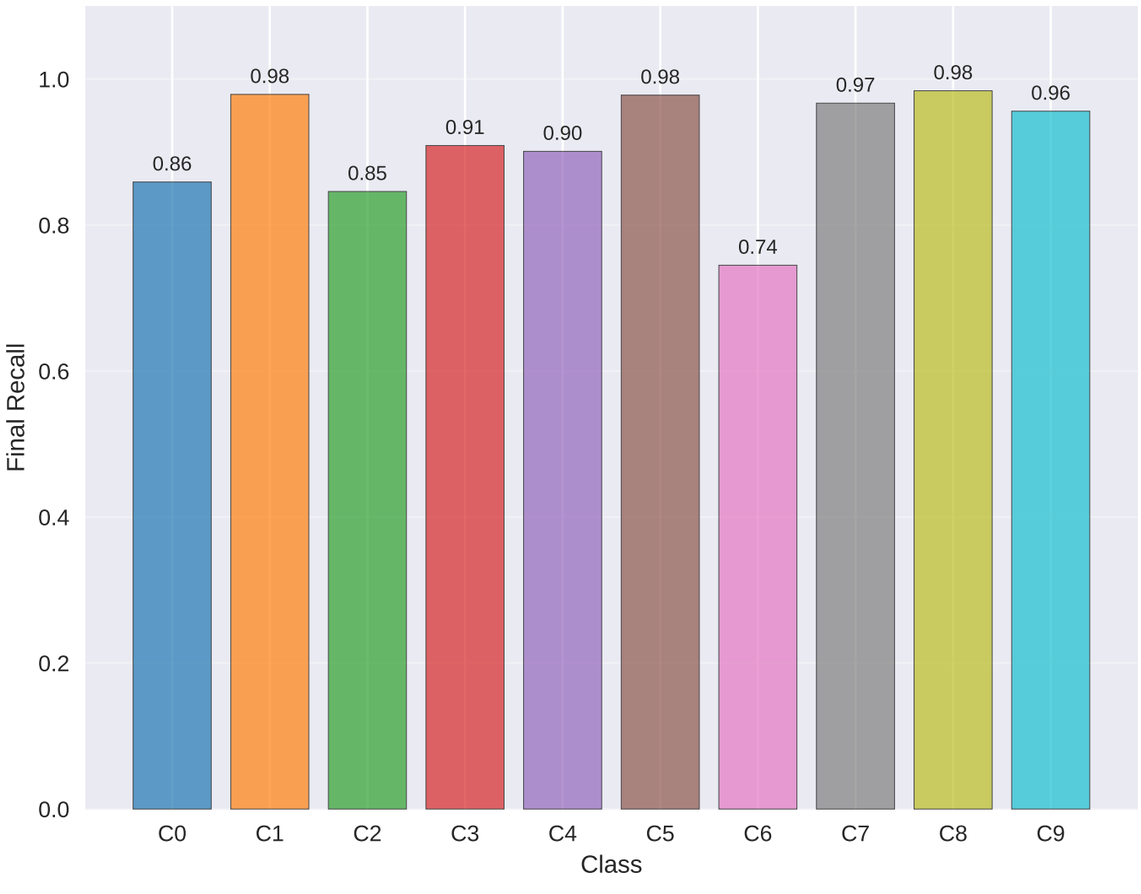
□ Precision Distribution per Class

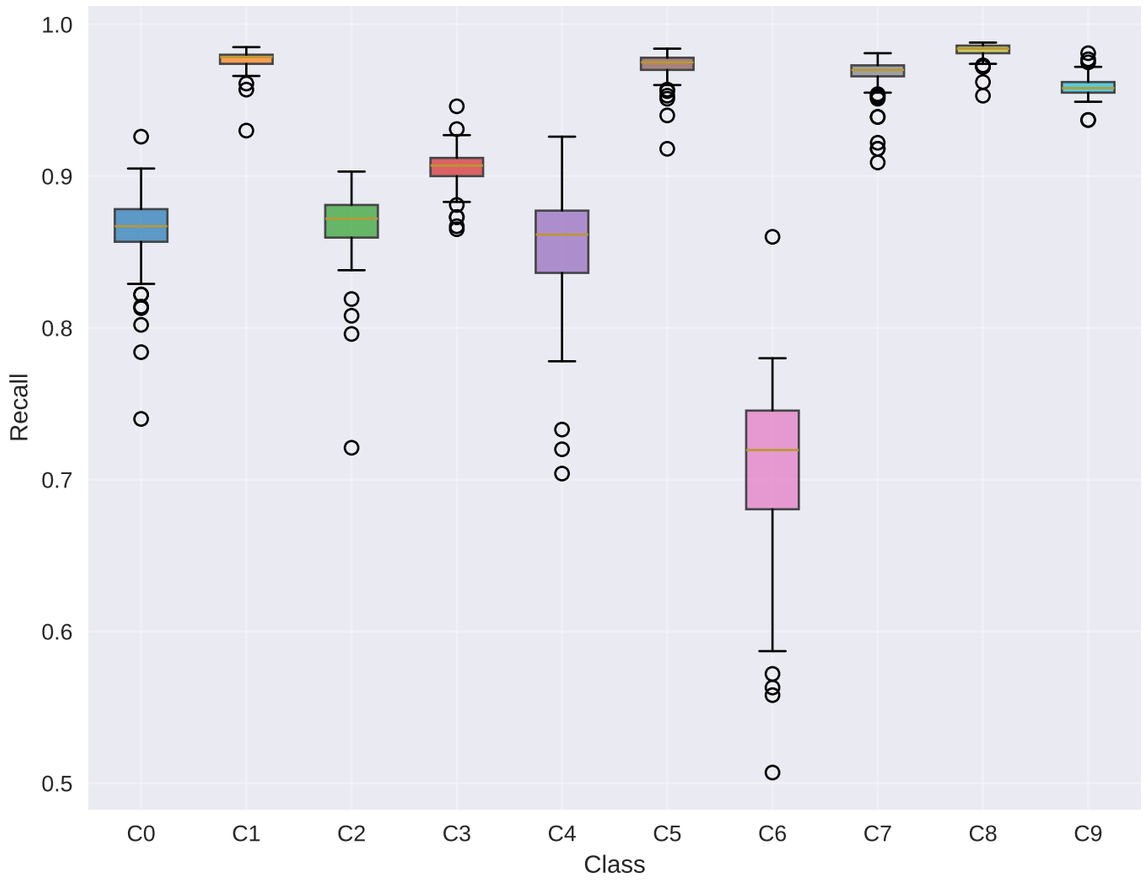


Per-Class Recall Analysis

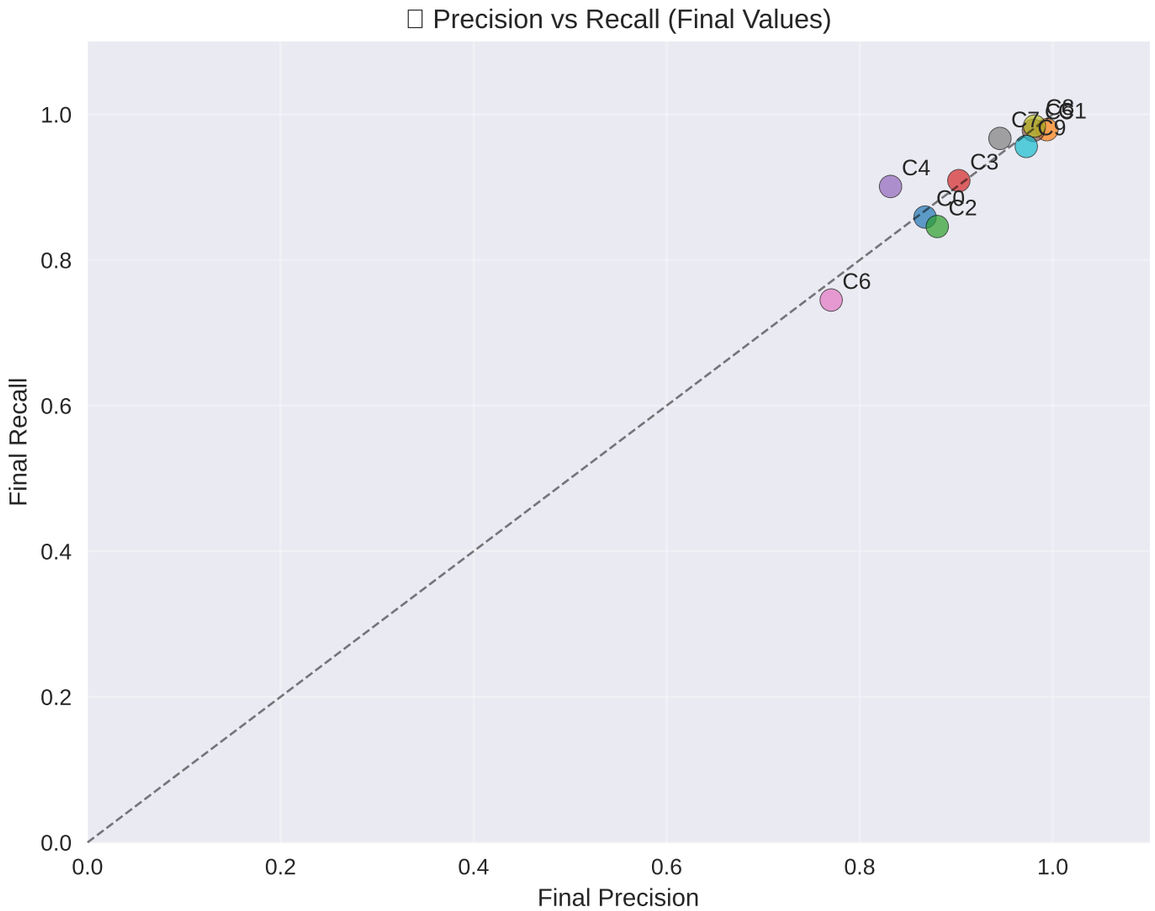




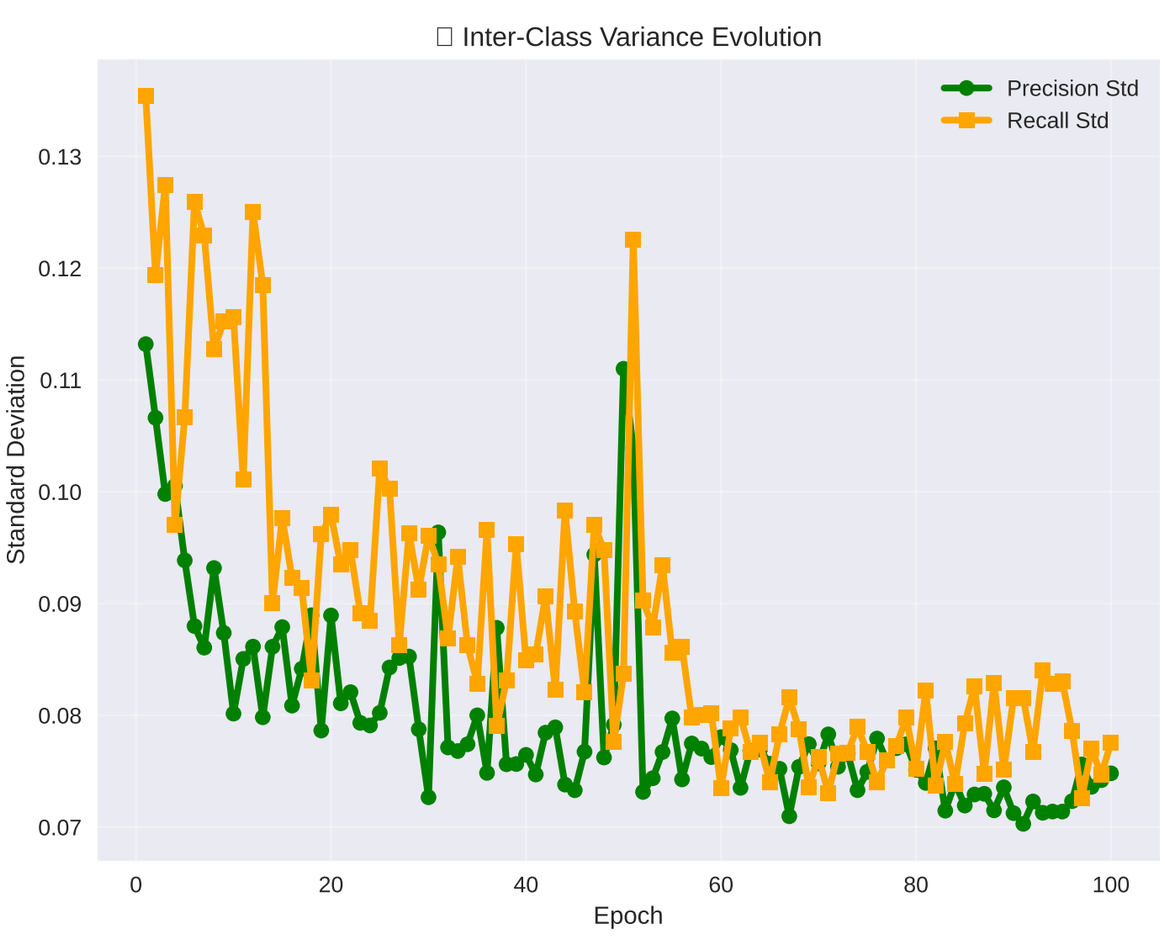


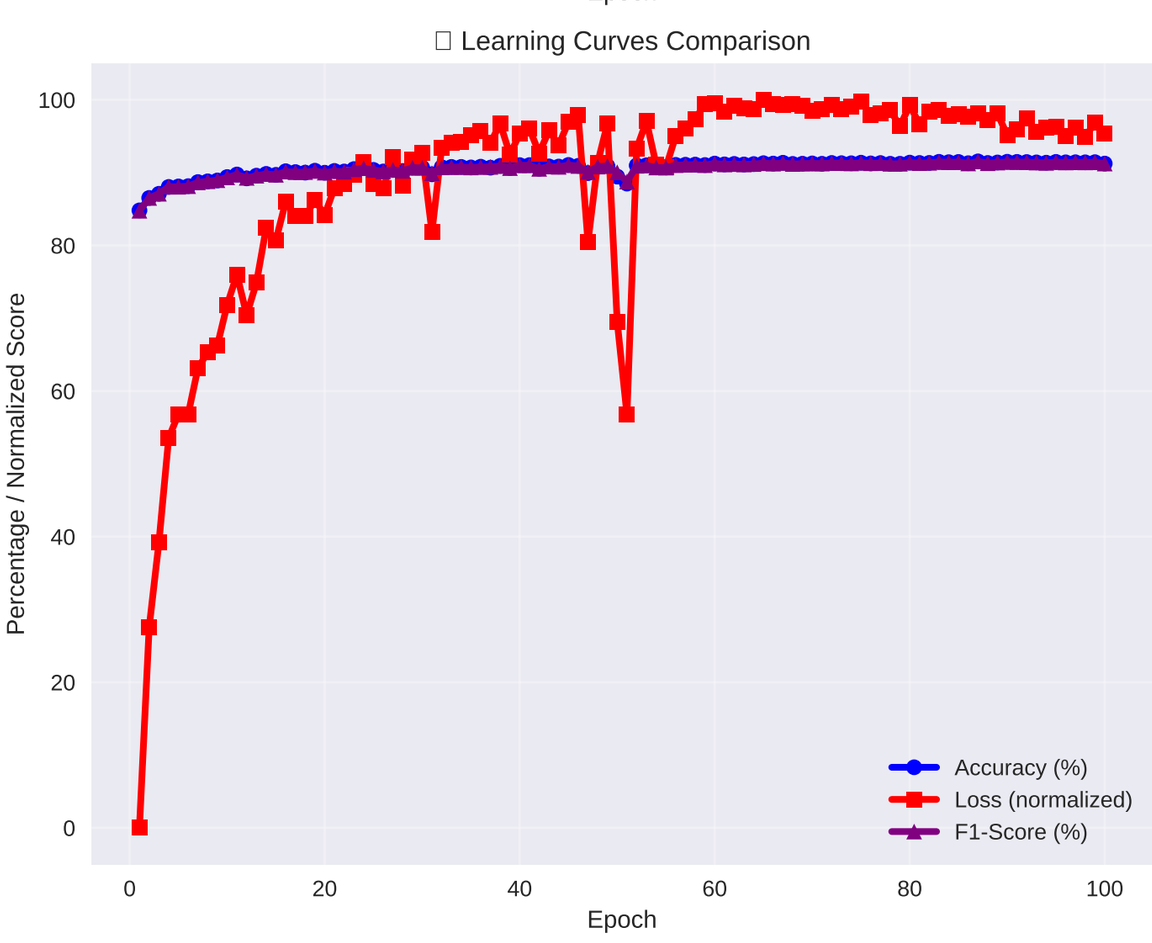


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-29 14:43:05
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.24%
• Loss: 2.5772
• Average Precision: 0.913
• Average Recall: 0.912
• Average F1-Score: 0.912

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.43%
• Riduzione Loss: +1.7321
• Stabilità Accuracy (CV): 0.0127

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.994)
• Peggiore Precision: Classe 6 (0.770)
• Migliore Recall: Classe 8 (0.984)
• Peggiore Recall: Classe 6 (0.745)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.868, R=0.859, F1=0.863
• Classe 1: P=0.994, R=0.979, F1=0.986
• Classe 2: P=0.880, R=0.846, F1=0.863
• Classe 3: P=0.903, R=0.909, F1=0.906
• Classe 4: P=0.832, R=0.901, F1=0.865
• Classe 5: P=0.980, R=0.978, F1=0.979
• Classe 6: P=0.770, R=0.745, F1=0.757
• Classe 7: P=0.945, R=0.967, F1=0.956
• Classe 8: P=0.981, R=0.984, F1=0.983
• Classe 9: P=0.973, R=0.956, F1=0.964

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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